

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (ECE) (Credit System) Degree Examinations

January 2023

20ECE63 – LINEAR ALGEBRA FOR SIGNAL PROCESSING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I

Marks BT* CO* PO*

1. a) Describe a column space and null spaces of the matrices:

$$A = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 & 0 & 3 \\ 1 & 2 & 3 \end{bmatrix}$$

6 L*3 1 1

- b) List the necessary properties of Vector spaces, subspaces.

8 L1 1 1

- c) Describe the subspace of R^3 spanned by:

i) The plane formed by vectors (b_1, b_1, b_3)

ii) All linear combinations of $v = (1, 4, 0)$ and $v = (2, 2, 2)$

iii) All vectors with positive components.

6 L3 2 1

2. a) Solve the given system of equations:

$$2x + y + z = 10$$

$$3x + 2y + 3z = 18$$

$$x + 4y + 9z = 16$$

7 L3 1 1

- b) Find the ranks of the given matrices.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix} C = \begin{bmatrix} 1 & c \\ c & c^2 \end{bmatrix}$$

6 L3 1 1

- c) Find the inverse of the given matrix by elimination

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}$$

7 L3 2 1

3. a) By row operations reduce each of the following matrix into its echelon matrices and write down the L and U matrices of the given matrices.

$$A = \begin{bmatrix} -1 & 3 & 5 \\ -2 & 6 & 10 \end{bmatrix} B = \begin{bmatrix} -1 & 3 & 5 \\ -2 & -6 & 7 \end{bmatrix}$$

8 L3 1 1

- b) Describe the subspaces that belong to R^2 and R^5 .

6 L2 1 1

- c) If V is the subspace spanned by $(1, 1, 1)$ and $(2, 1, 0)$ and find a matrix A that has V as row space. Find a matrix B such that V is the null space.

6 L3 1 1

Unit – II

4. a) Find the SVD of the given matrix :

$$A = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}$$

6 L2 3 1

- b) Explain orthogonal and orthonormal vectors with relevant examples.

6 L2 4 1

- c) With an example explain diagonalization of matrices.

8 L2 3 1

P.T.O.

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|-------|---|----|----|---|---|
| 5. a) | Explain the QR factorization using relevant examples. | 10 | L2 | 3 | 1 |
| b) | Discuss about the four fundamental subspaces and their respective dimensions considering an example. | 10 | L2 | 4 | 1 |
| 6. a) | Perform Gram-Schmidt orthogonalization to the given vectors: | | | | |
| | $a = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad b = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \quad c = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ | 10 | L3 | 4 | 1 |
| b) | Find the Eigen values and Eigen vectors of A , A^2 , A^{-1} and $A + 4I$.
Given | | | | |
| | $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ | 10 | L2 | 4 | 1 |

Unit – III

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|-------|---|----|----|---|---|
| 7. a) | Explain how DFT matrix can be interpreted as a linear transformation. | 10 | L2 | 5 | 1 |
| b) | Discuss the application of least squares with a suitable example. | 10 | L2 | 5 | 1 |
| 8. a) | Explain the application of DCT for data compression. | 10 | L2 | 5 | 1 |
| b) | Discuss the concepts of Curve Fitting with a suitable example. | 10 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
